

Sanjana Binte Siraj

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Education

Master of Science (MS) in Electrical and Computer Engineering

Aug 2025 – Present

Stony Brook University, State University of New York

CGPA: 3.13/4.00

- Notable Courses: ESE 561 (Theory of Artificial Intelligence), ESE 577 (Deep Learning Algorithms and Software, In Progress), ESE 590 (Practical Machine Learning and Artificial Intelligence, In Progress), ESE 568 (Computer and Robot Vision), ESE 599 (Graduate Research in Digital Image Processing), ESE 554 (Data Structures and Algorithms), ESE 555 (VLSI Design)

Bachelor of Science (BSc) in Computer Science & Engineering

Apr 2019 – Jul 2024

Bangladesh University of Engineering & Technology (BUET)

CGPA: 3.48/4.00

- Notable Courses: Machine Learning, Microprocessor, Microcontroller & Embedded Systems, Compiler, Computer Networking, Simulation & Modelling, Computer Security, Computer Graphics, Artificial Intelligence

Publication

Asif Ihtemadul Haque, Shamim-Al Mamun, Sanjana Binte Siraj, Tanvir R. Faisal, and Mahmuda Naznin. 2025. Leveraging Quantitative Analysis for Severity Assessment of Knee Osteoarthritis. In Proceedings of the 12th International Conference on Next Generation Computing, Communication, Systems and Security (NSysS '25). Association for Computing Machinery, New York, NY, USA, 16–23. <https://doi.org/10.1145/3777555.3777566>

Work Experience

Research Assistant

Jun 2024 – Jan 2025

Dept. of CSE, Bangladesh University of Engineering and Technology (BUET)

- Worked as a graduate research assistant under supervision of Dr. Mahmuda Naznin, Professor, Ex-Dept. Head, CSE, BUET.
- Keywords of the project: biomedical analysis, multimodal learning, vision transformers, applied deep learning.
- Project has been awarded **RISE Student Research Grant**

Project Intern

May 2023 – Jun 2023

eSRD-Lab & MySoft Limited

Full Stack Development

- Completed an internship project focused on developing a full-stack framework.
- Languages: Java, JavaScript; Frameworks: Angular, Spring Boot; Tools: GitHub; Database: MySQL

Research Experience

Graduate Research in Image Processing and Motion Deblurring

Jan 2026 – May 2026

Supervised by: Dr. Murali Subbarao, Professor, ECE, Stony Brook University

- Conducted graduate-level research in image restoration and motion deblurring.
- Developed Python-based experimental workflows for image processing and quantitative performance evaluation.
- Investigated image restoration techniques through implementation and comparative analysis.

Transfer Learning in Bone Morphology Study for Better Hip Fracture Assessment

Jun 2023 – Jun 2024

Co-Supervised by: Dr. Mahmuda Naznin, Professor, Ex-Dept. Head, CSE, BUET and Dr. Tanvir Faisal, Assistant Professor, ME, University of Louisiana at Lafayette

- Manuscript in submission
- Description: This research centers on predicting hip fracture risk by first predicting stress and strain distributions across the bone under various fall scenarios. These distributions and the density distribution are then projected onto images of the bone. The study then employs a modified vision transformer to classify images corresponding to different patient-fall case pairs based on their fracture risk, providing a comprehensive approach to fracture risk assessment.
- Project was awarded **BUET RISE Student Research Grant 2023**

Improvement on Hip Fracture Assessment Pipeline

Nov 2024 – Jan 2025

Co-Supervised by: Dr. Mahmuda Naznin, Professor, Ex-Dept. Head, CSE, BUET and Dr. Tanvir Faisal, Assistant Professor, ME, University of Louisiana at Lafayette

- Study in progress
- Description: This study's objectives are to replace the entire previous multi-stage pipeline with a single multimodal architecture that will directly take the qct images and textual metadata as input. Also different PEFT strategies will be employed to make the previous vision transformer more lightweight.

Technical Skills

- Languages: C, C++, Python, Java, JavaScript, x86 Assembly, MySQL
- Libraries: PyTorch, Keras, Scikit-Learn, Numpy, Pandas, SciPy
- Frameworks: Postgres, MySQL, React, Bootstrap, Angular, Spring Boot
- Technologies: Android Studio, HTML, LATEX, Git, Atmega32 Microcontroller, Logisim circuit simulator

Recent Projects

Radiology Report Generation Using Multimodal Large Language Model

Biomedical Image Analysis, Vision Transformer, Multimodal Learning, Deep Learning, Applied Machine Learning

- Project in progress
- Description: This study plans to offer a lightweight, comprehensive and clinically accurate architecture for generating chest X-ray reports. The proposed solution will integrate retrieval-based data augmentation techniques along with different fine-tuning strategies.

Detection of Adversarial Android Malware with Query-Efficient Transformation Strategies

Networks and Security, Large Language Models

- Project in progress
- Description: This study plans to offer a detection architecture of the malware produced through the latest and highly efficient malware generator [EvadeDroid](#). The proposed solution is going to use fine-tuned LLMs to detect relevant features for the detection.

Book Review Popularity Prediction using NLP techniques

Natural Language Processing

- Description: In this project we compare performance of different methods of feature preprocessing, both textual and non-textual features, on popularity prediction of book reviews.
- Links: [Documentation](#) [GitHub](#)

Watcher 24-7

Android App

- Description: Watcher 24-7 is an android app that allows active location sharing with emergency contacts. Pressing the phone's power button 3 times or pressing the big red emergency button on the app just once sends sms alerts to your emergency contacts and starts location sharing. The novelty of this project is that it included a design of a prototype of a wearable with a button which on pressing also starts the mentioned features.
- Links: [Documentation](#) [GitHub](#) [Media Coverage](#)

RivERO

Android App

- Description: RivERO is an android app that tracks real time river erosion rates and in risk or actively eroding zones of Bangladesh. It also sends out natural disaster warning SMS alerts based on user's registered location.
- Links: [GitHub](#)

Smart Mask

Arduino

- Description: Smart Mask a prototype of a smart mask with a fan for cooling and ventilation and sensors to detect the quality and temperature of air. The fan starts rotating whenever there's toxic substance like smoke etc. in the air or temperature is too high. It also sends the ppm values of toxic elements in the air to any phone connected to the mask via bluetooth.
- Links: [Documentation](#) [GitHub](#)

Achievements

- **BUET RISE Student Research Grant 2023** : Research Award for Outstanding Research Projects by Undergraduate Students
- **Champion**: Girls' Mobile Application Development Contest 2021 organized by Bangladesh Open Source Network – BdOSN
- **Champion**: BWCSE Project Showcasing 2021
- **2nd Runner Up**: MIST Inter University ICT Innovation Fest 2021 (Hackathon)
- **2nd Runner Up**: HackNSU_Season 3 (Hackathon)
- **Media Interview**: [Walton Laptop Campus Tech Talent Interview by Techshohor](#)

Other Activities

- **Head of Decoration Committee**, BUET CSE Fest 2023
- **2022 bdapps Campus Ambassador**, Bdapps, Robi